

Basic Code Analysis: Cross-Site Scripting (XSS) Vulnerability Audit

Duration : 30 Min.

Scenario

A security audit found an unsanitized `element.innerHTML` assignment vulnerable to DOM-based XSS attacks. Create a safe DOM renderer in `student_workspace/`.

Target File Location & Creation

File to Create/Update: `student_workspace/comments.js`

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Create 'comments.js' inside 'student_workspace/'. Use 'element.textContent' instead of 'element.innerHTML' to safely render user comments.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect the scenario input files in `student_workspace/`.
3. Create/update the file `student_workspace/comments.js` and populate it with valid parameters and configurations.
4. Save your changes (`Ctrl + S` or `Cmd + S`).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your work by running `python run.py` in the terminal before submitting.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	comments.js exists in student_workspace/ with valid JS code	3 Marks

TC2	Unsafe innerHTML property assignment removed	4 Marks
TC3	Safe textContent property assignment implemented	3 Marks

Total Score: 10 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.